R Niveau II

Solutions complètes — R Niveau II

Orientation : bases de données, extraction Excel, requêtes avancées et statistiques descriptives pratiques.

Jour 1 — SQL + Excel + requêtes avancées

Module 1 : Connexion et interrogation base

```
library(DBI)
library(RSQLite)
library(dplyr)
library(dbplyr)

con <- dbConnect(SQLite(), "business.db")

# Découverte
dbListTables(con)

# Requête distante
ventes_tbl <- tbl(con, "ventes")

ca_region <- ventes_tbl |>
  filter(date >= "2025-01-01") |>
  group_by(region) |>
  summarise(ca_total = sum(prix_net, na.rm = TRUE), .groups = "drop")

show_query(ca_region)
ca_region |> collect()
```

Module 2 : Extraction et normalisation Excel

```
library(readxl)
library(janitor)
library(tidyr)
library(stringr)
```

```
# Import propre
catalogue <- read_excel("catalogue.xlsx", sheet = "Produits") |>
  clean_names()

objectifs <- read_excel("objectifs.xlsx", sheet = "Objectifs", skip = 2) |>
  clean_names() |>
  mutate(
    objectif_ca = str_replace_all(objectif_ca, "[ €]", "") |>
      str_replace(",", ".") |>
      as.numeric()
  )

# Wide -> long
objectifs_tidy <- objectifs |>
  pivot_longer(
    cols = starts_with("m_"),
    names_to = "mois",
    values_to = "objectif_ca"
  )
```

Module 3 : Jointures et requêtes avancées

```
clients_tbl <- tbl(con, "clients")

base_analyse <- ventes_tbl |>
  left_join(clients_tbl, by = "client_id") |>
  collect() |>
  left_join(catalogue, by = "produit_id") |>
  mutate(montant_commande = quantite * prix_net)

# Produits vendus hors catalogue
produits_absents <- base_analyse |>
  distinct(produit_id) |>
  anti_join(catalogue |> distinct(produit_id), by = "produit_id")

# Top clients par région
top_clients <- base_analyse |>
  group_by(region, client_id) |>
  summarise(ca = sum(montant_commande, na.rm = TRUE), .groups = "drop") |>
  group_by(region) |>
  mutate(rang = min_rank(desc(ca))) |>
  filter(rang <= 5) |>
  arrange(region, rang)
```

Module 4 : Fenêtres et automatisation

```
library(purrr)

# Évolution mensuelle
evo_mensuelle <- base_analyse |>
  mutate(mois = as.Date(format(date, "%Y-%m-01"))) |>
  group_by(region, mois) |>
  summarise(ca = sum(montant_commande, na.rm = TRUE), .groups = "drop") |>
  group_by(region) |>
  arrange(mois, .by_group = TRUE) |>
  mutate(
    ca_mois_prec = lag(ca),
    var_mom = (ca - ca_mois_prec) / ca_mois_prec
  ) |>
  ungroup()

# Fonction paramétrée
extraire_kpi <- function(con, date_debut, date_fin, region = NULL) {
  req <- tbl(con, "ventes") |>
    filter(date >= date_debut, date <= date_fin)

  if (!is.null(region)) req <- req |> filter(region == !!region)

  req |>
    summarise(
      ca_total = sum(prix_net, na.rm = TRUE),
      nb_commandes = n_distinct(id_commande),
      panier_moyen = ca_total / nb_commandes
    ) |>
    collect()
}

# Import multi-fichiers robuste
safe_read_excel <- possibly(
  ~ read_excel(.x, sheet = "Data") |> clean_names(),
  otherwise = NULL
)

imports <- list.files("exports", pattern = "\\.xlsx$", full.names = TRUE) |
  set_names() |>
  map(safe_read_excel)

consolide <- imports |> compact() |> list_rbind(names_to = "source_fichier"
```

Jour 2 — Qualité + descriptif + restitution

Module 5 : Contrôles qualité

```
# Profil NA et cardinalité
profil_qc <- base_analyse |>
  summarise(across(
    everything(),
    list(na_n = ~ sum(is.na(.x)), n_distinct = ~ n_distinct(.x))
  ))

# Doublons de clé commande/date
doublons <- base_analyse |>
  count(id_commande, date, name = "n") |>
  filter(n > 1)

# Réconciliation base vs Excel
ca_base <- base_analyse |>
  mutate(mois = as.Date(format(date, "%Y-%m-01"))) |>
  group_by(mois, region) |>
  summarise(ca_base = sum(montant_commande, na.rm = TRUE), .groups = "drop"

reco <- ca_base |>
  full_join(objectifs |> rename(ca_excel = objectif_ca), by = c("mois", "re
  mutate(
    ca_base = coalesce(ca_base, 0),
    ca_excel = coalesce(ca_excel, 0),
    ecart = ca_base - ca_excel,
    ecart_pct = if_else(ca_excel == 0, NA_real_, ecart / ca_excel)
  )
```

Module 6 : Statistiques descriptives pratiques

```
# Indicateurs globaux
kpi_global <- base_analyse |>
  summarise(
    ca_total = sum(montant_commande, na.rm = TRUE),
    panier_moyen = mean(montant_commande, na.rm = TRUE),
    panier_mediane = median(montant_commande, na.rm = TRUE),
    iqr_panier = IQR(montant_commande, na.rm = TRUE),
    nb_clients = n_distinct(client_id)
  )

# Analyse segment
desc_segment <- base_analyse |>
  group_by(segment) |>
  summarise(
    nb_clients = n_distinct(client_id),
```

```
    ca_total = sum(montant_commande, na.rm = TRUE),
    panier_moyen = mean(montant_commande, na.rm = TRUE),
    panier_mediane = median(montant_commande, na.rm = TRUE),
    iqr_panier = IQR(montant_commande, na.rm = TRUE),
    .groups = "drop"
  ) |>
  mutate(part_ca = ca_total / sum(ca_total)) |>
  arrange(desc(ca_total))

# Série temporelle
serie_mensuelle <- base_analyse |>
  mutate(mois = as.Date(format(date, "%Y-%m-01"))) |>
  group_by(mois) |>
  summarise(ca = sum(montant_commande, na.rm = TRUE), .groups = "drop") |>
  arrange(mois) |>
  mutate(
    ca_mois_prec = lag(ca),
    var_mom = (ca - ca_mois_prec) / ca_mois_prec,
    cumul_annuel = cumsum(ca),
    mm3 = (lag(ca, 2) + lag(ca, 1) + ca) / 3
  )
```

Module 7 : Restitution

```
library(gt)
library(ggplot2)
library(readr)
library(openxlsx)

# Tableau de pilotage
tbl_segment <- desc_segment |>
  gt() |>
  fmt_currency(columns = c(ca_total, panier_moyen, panier_mediane), currenc
  fmt_percent(columns = part_ca, decimals = 1)

# Graphique tendance
p_ca <- serie_mensuelle |>
  ggplot(aes(mois, ca)) +
  geom_line(linewidth = 1.1, color = "#0984e3") +
  geom_point(size = 2, color = "#0984e3") +
  labs(title = "CA mensuel", x = "Mois", y = "CA") +
  theme_minimal()

# Graphique segment
p_seg <- desc_segment |>
  ggplot(aes(reorder(segment, ca_total), ca_total, fill = segment)) +
  geom_col(show.legend = FALSE) +
```

```
  coord_flip() +
  labs(title = "Contribution CA par segment", x = "Segment", y = "CA") +
  theme_minimal()

# Exports
write_csv(desc_segment, "outputs/desc_segment.csv")
write.xlsx(reco, "outputs/reco_base_excel.xlsx", overwrite = TRUE)
ggsave("outputs/ca_mensuel.png", p_ca, width = 9, height = 5, dpi = 150)
```

Projet final — Checklist de correction

Bloc	Vérification
Extraction	BDD + Excel chargés, colonnes normalisées, types cohérents
Qualité	NA, doublons, réconciliation traités et documentés
Requêtes avancées	Jointures correctes, indicateurs calculés sans doublonnage caché
Descriptif	KPI centraux + dispersion + segmentation + temporalité
Restitution	1 tableau + 2 graphiques + synthèse actionnable